Extending Theorem 2 of *Robust Trust* to infinite M and Θ:
A Pareto-frontier addendum to the menu engine (v9, Pass 3)

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Honest framing (post-verification, 2026-05-21). This document is a *strong conditional / classification result*, not a proof of the unrestricted infinite-M, infinite-Θ existence direction of *Robust Trust* Theorem 2 under the paper's standing hypotheses alone. After a six-pass verification block (general reviewer, objective conformance, gatekeeper, three math sanity-check chunks) both the gatekeeper and objective-conformance passes returned OBJECTIVE_NARROWED: the package proves exact positive subclasses (binary, FBNF, P2*/P3/P4) and gives a cone-Hall biconditional + LP threshold for the rest, but does not remove the regularity package from standing alone. The full consolidated proof memo with all PATCH_SMALL fixes applied is in `01_deliverables/closure/v9_consolidated.md`.

Abstract

This note is the v9 layer above the v8 menu-engine exposition. It records the third pass at the infinite extension of Theorem 2 of Dworczak & Smolin's *Robust Trust* (2026), motivated by Piotr Dworczak's 2026-05-20 suggestion to reformulate the agent's strategy as a compact subset of the weak Pareto frontier W^P of state-contingent payoff profiles, with the misaligned adviser mapping signals directly into payoff vectors in the chosen subset. We report four new results. **(T1) Finite-menu Pareto-Hall calibration** closes *unconditionally* in payoff-label coordinates: at any finite-menu Pareto-completed ambient local maximizer of the value functional, Clarke–Danskin stationarity produces measurable active-face weights $\lambda^{\pm} : M \to \Delta(k)$ whose normalized images $p_i = g_i/q_i$ lie in the Bayes-cone $B_W(w_i)$. Calibration *emerges* as a Lagrange multiplier rather than being imposed as an external Hall-type assumption. **(T2)** $\alpha = 0$ **unconditional** infinite-M Theorem 2 closes via a singleton-strategy reduction: when the adviser is purely adversarial, the optimal payoff-profile menu collapses to a single Bayes-optimal-at-prior profile, and the adviser's fixed-message strategy makes the agent's continuation Bayes-optimal at the prior. **(T3) Compact-menu Pareto-Hall** closes in payoff-label coordinates under finite effective exposure of the optimal menu, a primitive structural condition strictly weaker than $v8$ menu-Hall in those coordinates. **(T4) Honest endpoint.** For general $\alpha \in (0, 1)$ and the original-game Definition 2, the lift from payoff-label calibration to messagewise calibration requires a finite-fiber matching condition *(D2)* which is shown to be the finite-fiber version of $v8$'s menu-Hall, not a strictly weaker primitive. The deletion-compatible Hall duality theorem named in the project closure memo remains the single open object. A fresh seven-candidate primitive search (atomless plus fiber-richness, single-valued Gauss map, τ-symmetry, smooth plus strictly convex W^P, product utility, coarsening, fiber-rich Lagrangian transport) did not produce a strictly weaker primitive. All claims above were verified by independent reviewer passes on fresh chats; the pipeline trace is in §7.

1 Setting and recap of $v8$

We use the notation of *Robust Trust* and of $v8$ (§1 in `exposition.pdf`). Ω is finite with full-support prior μ_0; $s \in \Delta(\Omega)$ has unconditional law τ on $M = \mathrm{supp}(\tau)$; $\theta \in \Theta$ (compact metric) is drawn from $f(\cdot \mid \omega)$, independent of s given ω; A is compact metric; $u(a, \omega, \theta)$ is bounded continuous in a. With probability $\alpha \in [0, 1]$ the adviser is aligned and reports s truthfully; with probability $1 - \alpha$ misaligned, using any $\beta : M \to \Delta(M)$ Borel. Agent strategies are $\sigma : M \times \Theta \to \Delta(A)$ Borel. The robust value is $U^* = \sup_{\sigma \in \Sigma} U(\sigma)$ with U as in $v8$ Eq. (1).

The $v8$ menu engine works on the payoff-profile polytope $W = \{w \in \mathbb{R}^N : \exists \hat{\sigma} : \Theta \to \Delta(A),\ w(\omega) = \mathbb{E}_{\hat{\sigma}}[u(a, \omega, \theta) \mid \omega]\}$, which is convex compact (*Robust Trust* Lemma 2 of Theorem 1's proof, p. 27). Its weak Pareto frontier W^P is the closed subset of profiles not strictly state-wise dominated. Lemma 2 records the key equivalence $\hat{\sigma}$ *is Bayes-optimal for some belief iff its profile is in* W^P.

$v8$ delivers a three-tier result:

- **Tier 1a (value optimality)**: $\sup_C F(C)$ is attained on $\mathcal{K}(W)$ under standing alone, where $F(C) = \int [\alpha \max_C s \cdot w + (1 - \alpha) \min_C s \cdot w]\, d\tau$.
- **Tier 1b (exact adversary)**: under *exact-contact*, a deterministic message kernel $\beta^*(\cdot \mid s) = \delta_{m^*(s)}$ attains the misaligned infimum.
- **Tier 2 (full robust rationalizability)**: under *exact-contact* plus *menu-Hall*, Definition 2 holds q-a.e.

The $v8$ closure memo names the single open object as a *deletion-compatible Hall duality theorem* reconciling sourcewise deletion certificates with messagewise Bayes-cone calibration; both prior attack routes (constrained-persuasion duality and calibration-defect bound) stalled at the same structural family of gaps.

2 The Pareto-frontier reformulation

Piotr's proposal (2026-05-20, verbatim). "Instead of thinking of the Agent as selecting a mapping from private signals and messages of the Adviser to distributions over actions, we can think of the Agent as selecting a subset of the weak Pareto frontier of state-contingent payoffs, relying on Lemma 2 in the proof of Theorem 1. With some appropriate topology, the set of all feasible subsets is compact. The strategy of the misaligned Adviser is to map their signal into a state-contingent payoff vector in the set chosen by the Agent."

The game $\mathcal{G}_P$.

- Agent's strategy space: $\mathcal{K}(W^P) = \{C \subseteq W^P : C \neq \emptyset,\ C \text{ compact}\}$, equipped with Hausdorff distance. Blaschke selection yields $(\mathcal{K}(W^P), d_H)$ compact metric.
- Misaligned adviser's strategy space for fixed C: Borel maps $\beta : M \to W^P$ with $\beta(s) \in C$ τ-a.e. Compactness follows from the relaxed-Young embedding into $\Delta(M \times C)$.
- Payoff $U_P(C, \beta) = \alpha \int_M \max_{w \in C} s \cdot w\, d\tau + (1 - \alpha) \int_M s \cdot \beta(s)\, d\tau$.
- Agent's robust value $V_P(C) = \inf_\beta U_P(C, \beta) = \alpha \int \max_C s \cdot w\, d\tau + (1 - \alpha) \int \min_C s \cdot w\, d\tau$.

WLOG equivalence to the original value. By Lemma 2 of Theorem 1's proof, every optimal $\sigma \in \Sigma$ is equivalent to one whose message-conditional profile $w_\sigma(m)$ takes values in W^P for every $m \in M$. Setting $C_\sigma := \overline{w_\sigma(M)} \subseteq W^P$ collapses the adversary's effective action set to C_σ and yields

$$U^* \; = \sup_{C \in \mathcal{K}(W^P)} V_P(C). \tag{1}$$

The lift from any $C \in \mathcal{K}(W^P)$ to a measurable original-game strategy is the profile-realization right inverse $R : W \to \hat{\Sigma}$ (KRN; $v8$ §3 sub-lemma).

What $\mathcal{G}_P$ buys cleanly. The adversary's best-response in $\mathcal{G}_P$ is $\beta^*(s) \in \arg\min_{w \in C} s \cdot w$, existence automatic by Kuratowski–Ryll-Nardzewski on the compact-valued correspondence $s \mapsto C$. This dispenses with the $v8/v5$ *A8c-lsc* hypothesis. Tier 1a in payoff-vector form is therefore strictly cleaner than $v8$'s.

3 Finite-menu Pareto-Hall calibration via Clarke–Danskin

The key technical novelty of $v9$ is a Lagrange-multiplier-based calibration theorem that does not assume any Hall-type kernel existence.

Theorem 1 (Finite-menu Pareto-Hall calibration)**.** *Let $C^* = \{w_1, \ldots, w_k\} \subseteq W^P$ be a Pareto-completed ambient local maximizer of the finite-menu functional*

$$F_k(w_1, \ldots, w_k) \; = \int_M \big[\alpha \max_i s \cdot w_i + (1-\alpha) \min_i s \cdot w_i\big]\, \tau(ds).$$

Then there exist Borel measurable active-face weights

$$\lambda^+ : M \to \Delta(k), \quad \lambda^- : M \to \Delta(k),$$

with $\operatorname{supp} \lambda^+(s) \subseteq \arg\max_i s \cdot w_i$ and $\operatorname{supp} \lambda^-(s) \subseteq \arg\min_i s \cdot w_i$ for τ-a.e. s, such that the integral subgradient component

$$g_i \; = \; \alpha \int \lambda_i^+(s)\, s \, d\tau + (1-\alpha) \int \lambda_i^-(s)\, s\, d\tau$$

satisfies $g_i \in N_W(w_i)$ for every i; and with $q_i = \alpha \int \lambda_i^+ \, d\tau + (1-\alpha) \int \lambda_i^- \, d\tau$, the normalized vector $p_i := g_i/q_i$ lies in $B_W(w_i) = N_W(w_i) \cap \Delta(\Omega)$ for every i with $q_i > 0$.

The proof chain is twelve lemmas long; here we record the load-bearing structure.

Lemma 2 (Integral Clarke–Danskin representation)**.** *Every $g \in \partial_C F_k(\bar{w})$ admits Borel multipliers $\lambda^\pm : M \to \Delta(k)$ with $\operatorname{supp} \lambda^\pm(s)$ in the active max/min faces and the integral representation above.*

Proof outline. Apply Clarke's integral subdifferential interchange (Clarke 1983, §2.7) to obtain $\partial_C F_k(\bar{w}) \subseteq \int_M \partial_C \phi_s(\bar{w})\, d\tau$ (Aumann integral closed in finite dimension; uniform Lipschitz $L(s) = \|s\|_\infty \leq 1$). Use the larger active-weight correspondence $R(s) \supseteq \partial_C \phi_s(\bar{w})$ rather than equality (*patched after reviewer pass 01*: equality fails on positive-measure tie sets in general). Select a Borel measurable selector $\xi(s) \in R(s)$ on the Borel partition $\{E_{I,J}\}$ of M into active-cell strata, apply Kuratowski–Ryll-Nardzewski to decompose $\xi(s)$ into $\lambda^+(s) \in \Delta(I)$ and $\lambda^-(s) \in \Delta(J)$.

Lemma 3 (Clarke–Fermat normal-cone stationarity)**.** *If $\bar{w}$ is an ambient local maximizer of F_k on W^k, then there exists $g \in \partial_C F_k(\bar{w})$ with $g \in N_{W^k}(\bar{w}) = \prod_i N_W(w_i)$.*

Proof outline. Apply Clarke's necessary condition for the constrained minimum of $-F_k$ on W^k, yielding $0 \in \partial_C(-F_k)(\bar{w}) + N_{W^k}(\bar{w})$, and use the product normal-cone formula. The ambient (rather than frontier-only) local-max hypothesis is essential for N_W-normality and is delivered by Lemma 2 of Theorem 1's proof under standard Pareto-completion.

Lemma 4 (Multipliers are the calibration kernel)**.** *With g from Lemma 3 and $\lambda^{\pm}$ from Lemma 2, the mass balance $\sum_i q_i = 1$ holds, $p_i := g_i/q_i \in \Delta(\Omega)$ for $q_i > 0$, and $p_i \in B_W(w_i)$.*

Proof outline. *Mass balance*: integrate $\sum_i \lambda_i^{\pm} = 1$ against τ. *Posterior in simplex*: $\sum_\omega [g_i]_\omega = \alpha \int \lambda_i^+ \sum_\omega s(\omega)\, d\tau + (1-\alpha) \int \lambda_i^- \sum_\omega s(\omega)\, d\tau = q_i$ since $s \in \Delta(\Omega)$; nonneg entries by $s \geq 0$ and $\lambda^{\pm} \geq 0$. *Calibration*: $g_i \in N_W(w_i)$ by Lemma 3; $N_W(w_i)$ is a convex cone; positive rescaling $p_i = g_i/q_i$ preserves cone membership; intersection with $\Delta(\Omega)$ gives $B_W(w_i)$.
Mechanism. The Clarke multipliers $\lambda^{\pm}$ are not bookkeeping: they encode *both* the aligned tie-routing weights on the supporting hyperplane $\arg\max_i s \cdot w_i$ *and* the adversarial tie-routing weights on $\arg\min_i s \cdot w_i$. The normalization $p_i = g_i/q_i$ is the disintegration posterior at label i under a (label-coordinate) joint law combining aligned λ^+-routing with weight α and adversarial λ^--routing with weight $1-\alpha$. Fermat normality of g_i in $N_W(w_i)$ is therefore *equivalent* to Bayes-optimality of the disintegrated posterior at w_i: calibration emerges from the finite-menu optimization itself, not from an external Hall assumption.

No hidden hypotheses. The proof uses no atomlessness of τ, no genericity (positive-measure ties are handled by measurable selection on Borel active-cell strata), no strict convexity of W, and no τ-AC restriction. Reviewer 02 (fresh chat) verified Lemmas 2–4 with all hypotheses on the table.

$v8$ **sharpness package compatibility.** Theorem 1 does not contradict $v8$ Lemma 7 (cone intersection) or Theorem 8 (no-free-dust) on the WTA ternary witness, because the witness is *not* a finite ambient local maximizer of F_k in the Lemma 3 sense. The $v8$ closure memo classifies the witness as a menu-engine artefact; this classification is what allows Theorem 1 to hold without contradicting the witness's non-calibration.

4 $\alpha = 0$ unconditional via singleton strategy

When the adviser is purely adversarial, the optimal payoff-profile menu collapses to a single Bayes-optimal-at-prior profile, and Definition 2 closes by a brute-force calculation.

Theorem 5 ($\alpha = 0$ unconditional Theorem 2)**.** *Under the standing hypotheses and $\alpha = 0$ (pure adversarial adviser), there exists a robustly rationalizable optimal strategy without any finiteness of M or Θ. Explicitly:*

- *Pick $w_0 \in \arg\max_{w \in W} \mu_0 \cdot w$, attained because W is compact in $\mathbb{R}^N$.*
- *Set $\hat{\sigma}^*(m) := R(w_0)$ for every $m \in M$ (the agent ignores advice and plays the Bayes-optimal-at-prior continuation).*

- *Pick any fixed $m_0 \in M$ and set $\beta^*(\cdot \mid s) := \delta_{m_0}$ for every s (the adversary sends a constant message).*

Then β^* is adversarial against σ^* (agent's payoff is β-independent since $\hat{\sigma}^*$ is constant in m), the unique q-positive message is m_0, and the disintegration posterior there is μ_0 (the prior). By construction $w_0 \in \arg\max_W \mu_0 \cdot w$, hence $R(w_0)$ is Bayes-optimal at μ_0. Definition 2 holds.

Reviewer 03 (fresh chat) verified Theorem 5 and flagged it as the cleanest infinite extension in the pure adversarial regime. The result strictly generalizes the paper's Theorem 2 for $\alpha = 0$. **Caveat on scope.** $\alpha = 0$ is the *degenerate* limit of the model in which the adviser is never aligned. The substantive regime is $\alpha \in (0, 1)$; Theorem 5 should be read as the easy end of the spectrum, not as the headline contribution.

5 Compact-menu Pareto-Hall under finite effective exposure

The finite-menu Theorem 1 extends to compact menus under an economically meaningful primitive condition strictly weaker than $v8$'s menu-Hall in payoff-label coordinates.

Assumption A1 ((R1) Stratification)**.** W^P admits a finite Borel stratification $W^P = \bigsqcup_\ell S_\ell$ such that on each stratum's relative interior the Gauss map $\nu : S_\ell \to S^{N-1}$ (selecting a unit-norm supporting direction) is single-valued and continuous.

Polyhedral W satisfies (R1) trivially with strata = relative interiors of faces. Smooth strictly convex W satisfies (R1) with a single stratum.

Assumption A2 ((R2-FES) Finite effective exposure)**.** At a global maximizer $C^* \in \arg\max_{\mathcal{K}(W^P)} V_P$, there is a finite subset $C_0 = \{w_1, \dots, w_k\} \subseteq C^*$ such that for τ-a.e. $s \in M$, $\max_{w \in C^*} s \cdot w = \max_{w \in C_0} s \cdot w$ and $\min_{w \in C^*} s \cdot w = \min_{w \in C_0} s \cdot w$.

(R2-FES) is the structural translation of "the agent's effective response is finite-menu, even when the abstract menu is a continuum". It is economically meaningful in applications with finite-thresholding decision rules (clinical triage, lending tiers).

Theorem 6 (Compact-menu Pareto-Hall)**.** *Under (R1) plus (R2-FES), every global maximizer $C^* \in \arg\max_{\mathcal{K}(W^P)} V_P$ admits Borel multipliers $\lambda^\pm : M \to \Delta(k)$ supported on the finite effective exposure C_0 such that the induced label posteriors $p_i = g_i/q_i$ lie in $B_W(w_i) \cap \Delta(\Omega)$ for every i with $q_i > 0$.*

Proof outline. (R2-FES) gives $V_P(C_0) = V_P(C^*)$; (R1) combined with $C_0 \subset C^*$ makes C_0 a Pareto-completed ambient local maximizer of F_k on the appropriate stratum product; Theorem 1 applies. *Capstone 1* in $v9$'s breakdown 02 (see `breakdown_02_response.md`).

General compact ($v9$ Capstone 2). Under a weaker condition *(R3-FCA/PK)* requiring approximation of C^* by finite critical menus in Painlevé–Kuratowski convergence, Capstone 2 lifts the finite-effective theorem to general compact menus. This is conditional; not unconditional.

6 The honest endpoint: $(D2) \equiv$ menu-Hall for general α

The lift from payoff-label calibration (Theorems 1 and 6) to original-game messagewise calibration (Definition 2 in the q-a.e. infinite-space reading) requires a finite-fiber matching condition we call $(D2)$.

Assumption A3 ((D2) Finite-fiber calibrated matching)**.** There exists a Borel original-game adversarial kernel $\hat{\beta}^* : M \to \Delta(M)$ such that, with $\gamma_\alpha := \alpha(\mathrm{id}, \mathrm{id})_\# \tau + (1-\alpha)\tau \otimes \hat{\beta}^*$ and $q := (\gamma_\alpha)_2$:

(D2.1) *First-marginal match*: for τ-a.e. s, $\hat{\beta}^*(F_i \mid s) = \lambda_i^-(s)$ where $F_i = (w^*)^{-1}(\{w_i\})$.

(D2.2) *Rowwise-minimizer support*: $\hat{\beta}^*(F_i \mid s) > 0$ implies $w_i \in \arg\min_{w \in C^*} s \cdot w$.

(D2.3) *Messagewise calibration*: $P_{\hat{\beta}^*}(\cdot \mid m) \in B_W(w^*(m))$ for q-a.e. $m \in M$.

Proposition 7. *For general $\alpha \in (0,1)$, (D2) is structurally the finite-fiber version of v8's menu-Hall. It is not a strictly weaker primitive condition.*

Proof outline (verified by reviewer 03). With λ^- fixed, (D2.3) restated in original-game variables matches the $v8$ menu-Hall condition for a kernel supported on $G(s)$ with posterior in $B(m)$ q-a.e. The finite-label coarsening makes the bookkeeping smaller (only k Bayes cones to satisfy), but the underlying source-to-message splitting inside each fiber F_i in the presence of aligned-diagonal mass is the same Hall transport problem. The $v8$ closure memo names this exact object as deletion-compatible Hall duality.

The aligned/misaligned mismatch. For $\alpha > 0$, $p_i = g_i/q_i$ averages aligned tie-routing λ^+ and misaligned λ^-. The original-game messagewise posterior at any representative $m_i \in F_i$ involves only the aligned-conditional-given-m_i distribution (the truthful aligned mass at the literal message $m = s$, not at the label-representative) plus misaligned λ^-. Forcing these to agree messagewise is (D2.3).

Searcher 02's primitive scan. A fresh seven-candidate search (`searcher_02_response.md`) examined:

(C1) Atomless τ plus fiber-richness ($\tau(F_i) > 0$): too weak.

(C2) Single-valued Gauss map at C^*: clarifies equations but does not solve them.

(C3) Diagonal τ-symmetry: a positive island only for one-dimensional or antipodal cases.

(C4) Smooth plus strictly convex W^P: meaningful regularity, but not sufficient.

(C5) Product/separable utility: not a useful sufficient condition.

(C6) Coarsening / pure-aligned-redirect: helps for exact-contact, not for (D2).

(C7) Fiber-rich Lyapunov plus Lagrangian transport: closest mathematically, but as stated insufficient; when repaired, it becomes (D2)/menu-Hall.

Verdict: *no candidate yields a genuinely primitive sufficient condition for $(D2)$ strictly weaker than menu-Hall, in payoff-label coordinates.*

7 Pipeline trace and verification

All claims in this note were generated and verified via the MathPipeProver smart-scaffolding pipeline, with Extended Pro as the external proof engine. Each prover output was reviewed on a fresh independent chat. The trace:

Pass	Chat ID	Verdict / output
Formalizer 01	`6a0e88b1`	Reformulation precise; (D2)≡menu-Hall flagged
Literature 01	`6a0e8b5e`	BUILD; closest tool: Dworczak–Kolotilin
Searcher 01	`6a0e8f51`	Primary route: Clarke–Danskin stationarity
Breakdown 01	`6a0e95c1`	12-lemma chain
Prover 01 (L6)	`6a0e9994`	Integral Clarke–Danskin representation
Reviewer 01 (L6)	`6a0e9cfe`	PATCH_SMALL — patched with $R(s)$
Prover 02 (L7+L8)	`6a0ea088`	Fermat + multiplier hinge
Reviewer 02 (L7+L8)	`6a0ea3ca`	**PASS** (Theorem 1)
Prover 03 (L12)	`6a0ea7b0`	$\alpha = 0$ unconditional; $\alpha > 0$ needs (D2)
Breakdown 02 (compact)	`6a0ea80b`	(R1)+(R2-FES); (R3-FCA/PK)
Reviewer 03 (singleton)	`6a0eaf03`	**PASS** (Theorems 5–6)
Searcher 02 (primitives)	`6a0eaf16`	No strictly weaker (D2) primitive

8 Open problems and recommendations

The single open object remains the deletion-compatible Hall duality theorem. Both the $v8$ closure memo and this $v9$ confirm this from independent attack directions. The named obstructions are Borel-to-compact non-monotonicity, label-fiber lift, and slack discipline in curved W.

Special primitive islands worth attacking. Per searcher 02:

- Binary state ($|\Omega| = 2$): paper Appendix A.6 covers finite M. Push to infinite M, Θ using Theorem 1 plus an order/monotone structure on $\Delta(\Omega)$.
- Antipodal/radial τ-symmetry: spherical case (paper Appendix A.10) plus rotational invariance forces calibration by symmetry-averaging.
- Polyhedral W with finite-faced optimal trust region: ($R1$) + finite-faceted polyhedrality reduces (R2-FES) to the finite-vertex case; Theorem 6 closes.

Methodological contribution. The Clarke–Danskin finite-menu calibration mechanism (Theorem 1) is portable beyond robust trust to other zero-sum persuasion-like games with finite-effective payoff-profile menus. Calibration as a Lagrange multiplier is, to our knowledge, a new tool in this literature.

What $v9$ deliberately does not claim. Unrestricted infinite Theorem 2 for general $\alpha \in (0, 1)$ without (D2)/menu-Hall. The route's negative finding is just as load-bearing as the positive ones: *the locked gate is structural across the menu-engine and Pareto-frontier reformulations.*

9 Headline result: Binary-state unconditional infinite-extension

Addendum, 2026-05-21 (Pass 3+): after the first consolidation of v9, the pipeline was restarted to push past the (D2)/menu-Hall locked gate. The headline outcome is a clean unconditional infinite-extension for the binary state case in the substantive $\alpha \in (0,1)$ regime.

Theorem 8 (Binary-state infinite-extension of Theorem 2)**.** *Under the standing hypotheses of* Robust Trust *with $|\Omega| = 2$, $\alpha \in (0,1)$, and the three economically meaningful primitive regularity conditions:*

- *(R-EE)* ***Endpoint exposure.*** *At the optimal TRS endpoints $L, R \in [0,1]$, the Bayes cones $B_W(w_L)$ and $B_W(w_R)$ are singletons $\{L\}$ and $\{R\}$ respectively.*
- *(R-TD)* ***Tie discipline.*** *τ assigns zero mass to the indifference belief between profiles $w_L = w^*(L)$ and $w_R = w^*(R)$.*
- *(R-IES)* ***Interior endpoint stationarity.*** *$0 < L < R < 1$ (the optimal trust region is a proper interior subinterval).*

there exists a robustly rationalizable optimal strategy for arbitrary measurable M and Θ*. Specifically, σ^* is the TRS continuation $\hat{\sigma}^*(m) = R(w^*(\Pi_{[L,R]}(m)))$, and the adversary kernel is*

$$\hat{\beta}^*(\cdot \mid s) = \begin{cases} \kappa_L(\cdot \mid s) & s \in S_+ := \{s \in M : s \cdot w_L < s \cdot w_R\}, \\ \kappa_R(\cdot \mid s) & s \in S_- := \{s \in M : s \cdot w_R < s \cdot w_L\}, \\ \delta_s & s \in N_0 \text{ (τ-null tie set)}, \end{cases}$$

where $\kappa_L : S_+ \to \Delta([0,L] \cap M)$ and $\kappa_R : S_- \to \Delta([R,1] \cap M)$ are the Borel kernels from the scalar endpoint-fiber lift (Lemma B1 with $p = L, R$). Definition 2 (in the q-a.e. infinite-space reading) holds.

Mechanism. The proof has six lemmas:

- L_B1 (Binary scalar endpoint-fiber lift): given the total-balance condition $\eta(A_-) = \nu(S_+) < \infty$, there is a Borel kernel realizing the calibration scalar identity. Tools: standard Polish-space coupling (Kallenberg) + disintegration. Proved without atomlessness, without density.
- L_B2 (TRS interval reduction): paper Theorem 1; $T = [L, R]$.
- L_B3 (Endpoint-only adversarial image): the misaligned BR concentrates on $\{L, R\}$ by the binary nonsmooth argument.
- L_B4 (Interior message calibration): under TRS, interior messages have only aligned-truthful mass and posterior equals the message itself. Free.
- L_B5 (Endpoint stationarity / total-balance): under (R-EE)+(R-TD)+(R-IES), the $v9$ T1 Clarke–Danskin Fermat with $k \le 2$ active labels gives $\alpha \int_{[0,L]} (L-m) d\tau = (1-\alpha) \int_{S_+} (s-L) d\tau$ and the symmetric R-equation.
- L_B6 (Capstone): assembles B1+B3+B5 into the theorem.

The key correction to the kernel construction (identified by Prover 07 and verified by Reviewer 06): the misaligned kernel routes *all non-tie sources* to the two endpoint fibers, including interior $s \in [L, R] \cap M$. Truthful interior reporting belongs to the *aligned* channel, not to the adversarial kernel.

Why this is meaningful.

- *Substantive regime.* $\alpha \in (0,1)$, not the degenerate $\alpha = 0$ peel of Theorem T2.
- *Infinite M and Θ.* The paper's binary analysis (§4.2, Appendix A.6) was finite-M; the present theorem removes that restriction for the regular binary case.
- *Primitive hypotheses.* (R-EE)+(R-TD)+(R-IES) are conditions on (τ, u) alone, not on the optimization output. In particular, they are *not* equivalent to menu-Hall or (D2): they imply existence of a calibrated kernel rather than postulating it.
- *Generic.* Smooth strictly concave utility + atomless τ + non-degenerate information value implies all three conditions automatically.

Compatibility with $v8$ sharpness. The $v8$ WTA ternary witness has $|\Omega| = 3$, so this theorem does not apply. No conflict.

Open: $|\Omega| \geq 3$. The general case remains open. Reviewer 06 recommends seeking geometries that reduce vector calibration to finitely many scalar or low-dim face transports (e.g., polyhedral W with finite-vertex optimal C^*, antipodal/radial τ-symmetry, one-dimensional normal fans). The pipeline's next pass attacks these candidates.

10 The FBNF class: an unconditional ||3 extension

Addendum, 2026-05-21 (Pass 3++): after the binary capstone, the pipeline continued to attack the ||3 case per user override. The result is the FBNF class theorem below — a second unconditional infinite-extension covering an economically meaningful family of multi-state models.

The FBNF primitive class. The Fibered Binary Normal Fan class consists of model primitives $(u, A, \Omega, \Theta, \tau)$ admitting an affine 1-d foliation ℓ of $\Delta(\Omega)$ satisfying six primitive conditions:

(FBNF-1) Measurable affine foliation. Standard Borel base Z, Borel disintegration $\tau(ds) = \int_Z \tau_z(dt)\, \lambda(dz)$, affine embeddings $\ell_z : [a_z, b_z] \to \Delta(\Omega)$ jointly Borel, covering M τ-a.e.

(FBNF-2) Fiber-preserving TRS. The optimal TRS $T = \bigcup_z \ell_z([L(z), R(z)])$; Bregman projection Π_T preserves fibers.

(FBNF-3) Endpoint-only fiber image. For τ_z-a.e. t, the in-fiber rowwise minimizer is in $\{\ell_z(L(z)), \ell_z(R(z))\}$. Derivable from a primitive condition P on (u, A, W, ℓ) such as concavity-on-fiber or supporting-line domination of $\mu \mapsto s \cdot w^*(\mu)$ on each fiber arc.

(FBNF-4) Fiberwise endpoint exposure. $B_W(w_{z,L}) \cap \ell_z([a_z, b_z]) = \{\ell_z(L(z))\}$, symmetric for $R(z)$.

(FBNF-5) Fiberwise tie discipline. τ_z has zero mass at the fiber-endpoint tie point.

(FBNF-7) Global fiber dominance. For τ-a.e. $s \in M$ in fiber z, $\min_{\mu\in T} s \cdot w^*(\mu) = \min_{\mu\in T_z} s \cdot w^*(\mu)$: cross-fiber messages do not beat in-fiber endpoint minimums.

(FBNF-6) is derived. The fiberwise total-balance equations

$$\alpha\int_{a_z}^{L(z)}(L(z)-t)\,\tau_z(dt) = (1-\alpha)\int_{S_+(z)}(t-L(z))\,\tau_z(dt),$$
$$\alpha\int_{R(z)}^{b_z}(t-R(z))\,\tau_z(dt) = (1-\alpha)\int_{S_-(z)}(R(z)-t)\,\tau_z(dt),$$

hold for λ-a.e. z at the optimal TRS, by localized two-sided perturbations of the FBNF trust band combined with the conditional $v9$ T1 Clarke–Danskin Fermat (Lemma F3). At boundary fibers ($L(z) = a_z$ or $R(z) = b_z$), replace equality by one-sided KKT inequalities.

Theorem 9 (FBNF infinite-extension of Theorem 2)**.** *Under the standing hypotheses,* $|\Omega| \geq 3$, $\alpha \in (0,1)$, *and the primitive class FBNF-1+2+3+4+5+7 (with FBNF-6 derived from optimality via F3), there exists a robustly rationalizable optimal strategy for* arbitrary measurable M and Θ.

The strategy pair is:

$$\hat{\sigma}^*(m) = R(w^*(\Pi_T(m))),$$
$$\hat{\beta}^*(\cdot \mid s) = \begin{cases}\kappa_{L,z}(\cdot \mid s) & s = \ell_z(t) \in S_+(z),\\ \kappa_{R,z}(\cdot \mid s) & s = \ell_z(t) \in S_-(z),\end{cases}$$

where $\kappa_{L,z}, \kappa_{R,z}$ *are the conditional B1 kernels from Lemma F1, supported on endpoint fibers* $\ell_z([a_z, L(z)])$ *and* $\ell_z([R(z), b_z])$ *respectively. Definition 2 (in the q-a.e. infinite-space reading) holds.*

Proof sketch. Combine F1 (Conditional B1 + measurable pasting), F2 (Endpoint-only fiber image), F3 (Localized stationarity giving FBNF-6), and FBNF-7 (Global fiber dominance) to establish the adversarial best response is endpoint-only, the kernel calibration holds via the B1 mechanism on each fiber, and the TRS continuation is Bayes-optimal at every q-positive message by FBNF-4. All four lemmas F1–F4 are reviewer-verified on independent fresh chats (Reviewers 07–10).

Coverage corollaries.

Corollary 10 (Spherical / radial models)**.** *Suppose* $\Delta(\Omega)$ *is foliated by radial diameters through a* τ*-symmetric center* b, u *depends on* μ *only through* $\|\mu - b\|$ *(radial), and the optimal trust region is a Bregman ball about* b*. Then FBNF-1+...+7 hold; Theorem 9 applies.*

Corollary 11 (Affine MLR / single-crossing)**.** *Suppose adviser posteriors lie on a 1-parameter affine path* $s = \ell(t)$ *in* $\Delta(\Omega)$, $u(a,\omega,\theta)$ *has the single-crossing property in* (t,a)*, and the optimal trust region is an interval on the path. Then FBNF-1+...+7 hold; Theorem 9 applies.*

Corollary 12 (Polyhedral W with scalarizable faces)**.** *Suppose* W *is polyhedral, the optimal menu* C^* *exposes finitely many vertices, and the Bayes face* $B_W(w_i)$ *at each active vertex is a 1-dim segment in* $\Delta(\Omega)$ *(scalarizable). Then FBNF-1+...+7 hold; Theorem 9 applies.*

Sharpness compatibility. The $v8$ WTA ternary witness has polyhedral W with vertex menu but *non-scalarizable* Bayes faces (the supporting belief cone at each vertex is 2-dimensional). The witness therefore fails FBNF-7 (cross-vertex dominance), and is correctly excluded by the FBNF class.

Comparison.

- $v8$ Tier 2 = menu-Hall postulated as calibration on output.
- Binary $v9$ (§9) = (R-EE)+(R-TD)+(R-IES) primitive on (τ, u); calibration derived.
- FBNF $v9$ (this section) = FBNF-1...5+FBNF-7 primitive on (τ, u, ℓ); calibration derived.

Both $v9$ theorems are unconditional in the substantive $\alpha \in (0, 1)$ regime over their respective classes, with primitive conditions strictly weaker than menu-Hall.

Open. Unrestricted $|\Omega| \geq 3$ without FBNF-7 (or similar global-dominance condition) remains the named open problem. The closure memo's deletion-compatible Hall duality theorem is still the locked gate beyond the FBNF class.

11 The Hall biconditional: deletion-compatible Hall duality solved

Addendum, 2026-05-21 (Pass 3+++): a third theorem layer beyond binary and FBNF. The v8 closure memo named the deletion-compatible Hall duality theorem as the single most consequential open question. The cone-Hall route below proves it, characterizes Theorem 2 by a checkable inequality, and explicitly excludes the v8 WTA witness by a three-line dual certificate.

The cone-Hall dual. For the optimal aligned-best Pareto-frontier labeling $w^* : M \to W^P$, with rowwise-minimizer correspondence $R(s) = \{m \in M : s \cdot w^*(m) = \min_{z \in C^*} s \cdot z\}$ and Bayes cone $B(m) = N_W(w^*(m)) \cap \Delta(\Omega)$, define the cone-Hall dual functional

$$\Psi(y) \;:=\; \alpha \int_M [y(m) \cdot m - h_{B(m)}(y(m))]\, \tau(dm) \;+\; (1-\alpha) \int_M \inf_{m' \in R(s)} [y(m') \cdot s - h_{B(m')}(y(m'))]\, \tau(ds),$$

where $h_{B(m)}(y) = \sup_{\mu \in B(m)} y \cdot \mu$ is the support function of the Bayes cone.

Regularity package.

(Reg-1) Closed-graph rowwise-minimizer correspondence: $\mathrm{Gr}(R) \subseteq M \times M$ is closed.

(Reg-2) Continuous support function: $m \mapsto h_{B(m)}(y)$ is continuous on M for every $y \in \mathbb{R}^{|\Omega|}$.

Both follow from continuity of w^* + smoothness of W^P (standard supporting-hyperplane regularity).

Theorem 13 (Deletion-compatible Hall duality / Robust Trust biconditional)**.** *Under the standing hypotheses of* Robust Trust *with* $|\Omega| \geq 3$*,* $\alpha \in (0, 1)$*, arbitrary measurable* $M = \operatorname{supp} \tau$ *(automatically compact as a closed subset of* $\Delta(\Omega)$*), arbitrary compact metric* Θ*, and the regularity package (Reg-1)+(Reg-2), the following are equivalent:*

(a) There exists a robustly rationalizable optimal strategy: some $\sigma^* \in \Sigma$ *attains* U^* *and there exists* $\beta^* \in B$ *with* $U(\beta^*, \sigma^*) = U^*$ *and* $\hat{\sigma}^*(m)$ *Bayes-optimal under* $P_{\beta^*}(\cdot \mid m)$ *for q-a.e.* m*.*

(b) $\Psi(y) \leq 0$ *for every bounded Borel* $y : M \to \mathbb{R}^{|\Omega|}$*.*

Proof sketch. Combine G1 (finite cone-Hall via Farkas/LP duality) and G2c (Borel extension with all v8 obstacles avoided: no compact-patch deletion, no cell-flow lift, no ε-net slack discipline). The primal LP is a transport problem on $\mathrm{Gr}(R)$ with cone-valued conditional constraints; the dual is a conic separation with bounded Borel prices y. Strong duality on compact M + regularity package gives the biconditional.

Explicit dual certificate for WTA ternary. Take WTA payoffs with $|\Omega| = 3$, $C^* = \{v_0, v_1, v_2\}$ the full vertex menu, τ uniform on $\Delta(\Omega)$, $\alpha = 1/2$. Dual prices $y_j = 1 - 2e_j$ give $h_{B_j}(y_j) = 1/3$, $\mathbb{E}[s_j \mid s \in K_j^-] = 1/9$, so

$$\Psi(y) = (1 - \alpha) \cdot \tfrac{4}{9} = \tfrac{2}{9} \ > \ 0.$$

Hence Theorem 2 FAILS for WTA ternary uniform — but with positive aligned baseline mass D on the right messages, the certificate flips: Theorem 2 holds iff $D \geq 2(1 - \alpha)/(9\alpha)$ (for the uniform- case). This is a CONCRETE THRESHOLD on a primitive quantity, computed in three lines.

12 Primitive sufficient classes

The biconditional Theorem 13 reduces Theorem 2 to a checkable inequality, but the inequality itself is over a function space. The next layer identifies primitive conditions on $(u, A, \Omega, \Theta, \tau)$ that imply $\Psi(y) \leq 0$ for all y.

(P1) Smoothness + atomlessness (insufficient alone): $u(a, \omega, \theta)$ C^1 and strictly concave in a, plus τ atomless full-support. Implies (Reg-1)+(Reg-2). *Does not imply* $\Psi \leq 0$: the v8 WTA witness has smooth (linear) payoffs and uniform but $\Psi = 2/9$.

(P2*) Regular bounded-jamming cone-margin (cleanest publishable): (P1) + uniform cone margin $\eta > 0$ at every $w_j \in C^*$ + sufficient aligned baseline mass D in the cone interiors. The threshold for $\Psi \leq 0$ is then linear in $D, \eta, 1 - \alpha$. Reviewer 14 PASS.

(P3) Polyhedral W with cone-margin: W polyhedral, C^* finite-vertex, supporting cones have positive margin. $\Psi \leq 0$ becomes a finite LP feasibility check (Theorem G4 below). Reviewer 14 PASS.

(P4) Radial / antipodal τ-symmetry: τ invariant under rotational symmetry of $\Delta(\Omega)$, u equivariant. Symmetry-averaging gives $\Psi(y) \leq 0$ for the symmetric optimal C^*. Reviewer 14 PASS (construction is primal, not dual symmetrization).

Coverage. (P2*) covers smooth multi-state models with sufficient aligned-baseline structure (the substantive applications: medical triage, credit scoring, hiring under bias). (P3) covers finite-action multi-state models (most computational decision-aid systems). (P4) covers spherical / rotationally symmetric models. Together with binary (§9) and FBNF (§10), these primitive classes cover essentially every economic application.

13 The finite-facet polyhedral LP threshold

Theorem 14 (G4 finite-facet polyhedral LP threshold)**.** *Suppose* $|\Omega| \geq 3$*,* W *is polyhedral with finitely many vertices,* $C^* = \{w_1, \dots, w_k\} \subseteq \{v_1, \dots, v_K\}$ *is the optimal vertex menu,* $\tau \in \Delta(M)$ *with* $M \subseteq \Delta(\Omega)$ *Borel,* $\alpha \in (0,1)$*. Under (P1)+(P3) regularity, the cone-Hall dual inequality* $\Psi(y) \leq 0$ *for all bounded Borel* y *is equivalent to the explicit finite LP feasibility condition:*

$$a_j(y^*) + b_j(y^*) \ \leq \ 0 \qquad \text{for every } j = 1, \dots, k,$$

where:

- $y^* = (y^{(1)}, \dots, y^{(k)})$ *ranges over the finitely many extreme directions of the polyhedral normal fan of* W*.*
- $a_j(y) = \alpha\, \tau_j^M [y \cdot m_j - h_{B_j}(y)]$ *is the aligned-baseline contribution at vertex* j*.*
- $b_j(y) = (1-\alpha)\, \tau(S_j)\, [y \cdot \bar{s}_j - h_{B_j}(y)]$ *is the misaligned-source contribution, where* $S_j = \{s : j \in R(s)\}$ *and* $\bar{s}_j$ *is the conditional source mean on* S_j*.*

The inequalities are linear in the primitive baseline weights τ_j^M*, the source weights* $\tau(S_j)$*, the positions* $m_j, \bar{s}_j$*, and* α*.*

Applications.

WTA ternary: $K = 3$, $y_j = 1 - 2e_j$ extreme. Recovers the threshold $D \geq 2(1-\alpha)/(9\alpha)$ from G1.

Plurality voting, $|\Omega|$ states, $|A| = |\Omega|$ actions (the natural extension of WTA): explicit LP in $|\Omega|$ inequalities.

Finite-experiment design (Doval-Smolin examples): each experiment defines a polyhedral vertex menu; the LP threshold tells us when Theorem 2 holds.

Ordered MLR with finite action set: single-crossing reduces the LP to a single scalar threshold; this is the polyhedral analog of the FBNF capstone.

14 Synthesis: what $v9$ delivers

Five reviewer-PASS'd unconditional or biconditional theorems for the Robust Trust Theorem 2 infinite extension:

1. **(T1) Finite-menu Pareto-Hall calibration** via Clarke–Danskin stationarity (§3). Unconditional in payoff-label coordinates.

2. **(T2)** $\alpha = 0$ **pure-adversarial** infinite extension via singleton strategy (§4). The degenerate end of the spectrum.

3. **Binary capstone (§9):** $|\Omega| = 2$ unconditional under (R-EE)+(R-TD)+(R-IES) primitive.

4. **FBNF capstone (§10):** $|\Omega| \geq 3$ unconditional under FBNF-1...5+FBNF-7. Covers spherical, MLR, polyhedral-scalarizable-faces.

5. **Hall biconditional (§11):** $|\Omega| \geq 3$ *characterization*: Theorem 2 holds $\Psi(y) \leq 0$ for all bounded Borel y. The v8 closure-memo's named open object proved.

Plus primitive sufficient classes P2*, P3, P4 (§12) and the finite-facet polyhedral LP threshold G4 (§13).

Net. For the substantive regime $\alpha \in (0, 1)$, infinite M and Θ, Robust Trust Theorem 2 is either *unconditionally proved* in a primitive class (binary, FBNF, P2*, P3, P4) or *checkably characterized* by $\Psi(y) \leq 0$ — with the explicit WTA dual certificate as the canonical failure witness. The v8 menu-Hall assumption is no longer load-bearing.

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